SOHO Network Configuration on PA-220

WRITTEN BY

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Purpose

The main purpose of this lab is to use an unconfigured Palo Alto Firewall to set up a SOHO network using the web GUI. To do this, we must follow Palo Alto's "Setting up the PA-200 for Home and Small Office" guidelines. When the firewall is fully configured, the device should be able to access the internet through its ethernet connection to the firewall with the Wi-Fi turned off, which is the end goal.

Background information on lab concepts

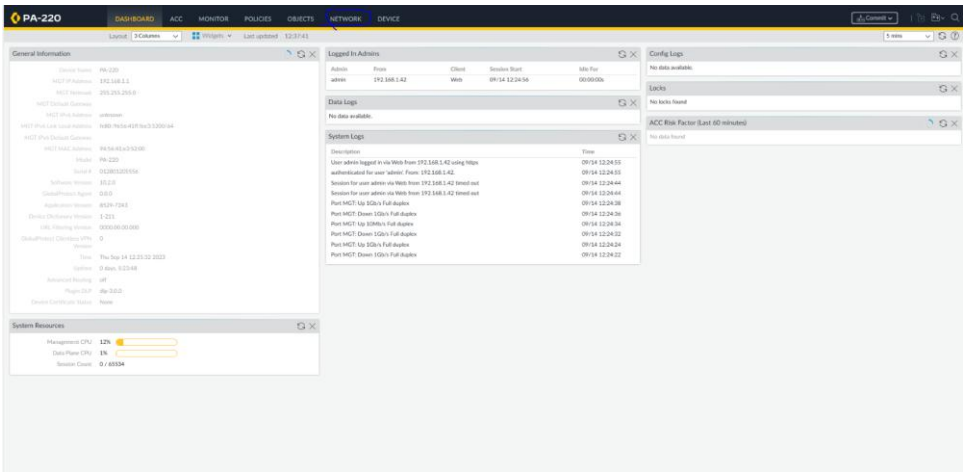
A SOHO (Small office and Home Office) network are local LANs (Local area Networks) that are usually connected to under 10 devices. Most of the time it is used in connecting multiple devices between a corporate's network and the LAN's network. It is also beneficial to set these up so company employees can access the company's network from home and would therefore be able to have the resources to work from home. However, by directly connecting your network to the corporate's network, if malicious code were to accidentally be sent by the end user, it might end up infecting the corporate's network as well. A firewall can be used to filter out malicious data and cyber-attacks such as DOS. By using a firewall to be able to access the internet or corporate network from your SOHO network, it allows your home devices to access the corporate network without the risk of spreading malicious code into the network. Part of the reason why Palo Alto firewalls are so expensive, is the different protocols and policies that these firewalls follow, meaning a different way to enforce these policies. Palo Alto's firewalls can extract information about the application before they allow it to run and can run without slowing down your network as much as other firewalls. To access the internet through your palo alto firewall, you need a switch or modem connected to the internet to be connected to the firewall for the firewall to actually be able to connect you to the internet. Specifically in Palo Alto Firewalls, a security zone must be applied to each interface. A security zone is a way to group together different physical or virtual interfaces on the firewall to filter what traffic can pass through that interface. The purpose of these zones is to allow the Palo Alto firewall to configure specific security policies to each zone instead of each firewall. To allow a firewall to access the internet and filter traffic, a network zone must be configured on each interface, along with a VLAN configured to each interface. A VLAN is also a way of grouping different firewalls together, many security teams can mass configure firewalls through their grouping to a specific VLAN.

Lab summary:

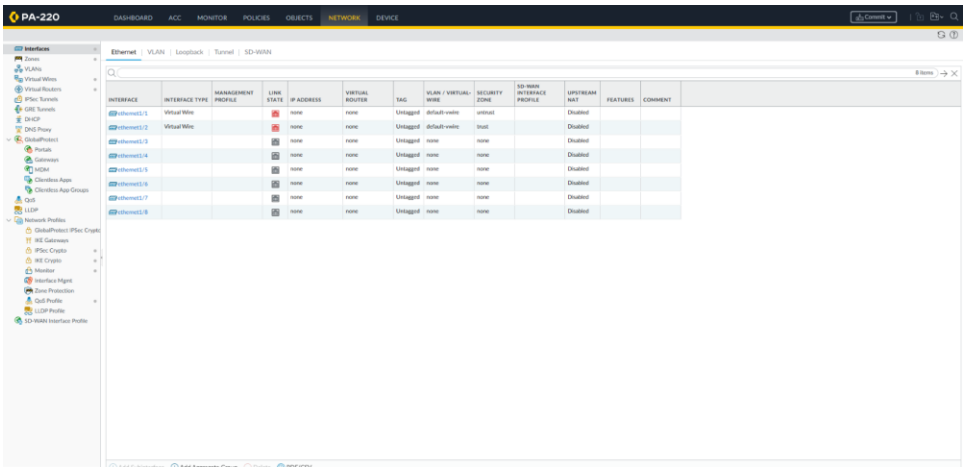
Entering the web GUI, under Network > Zones, we configured 3 zones. Trust-L2, Trust-L3, and Untrust-L2. Next, we configured the physical interfaces. We set eth1/1 to be layer 3, a virtual router of default, a security zone of untrust-L3, and an automatically assigned p address from dhcp. next, we created a VLAN object called "VLANMAN" assigned to the interface of vlan.10. We then assigned the VLANMAN interface to eth1/2, eth1/3, and eth1/4 with a security zone of

trust.L2. We now needed to assign an IP address to the vlan to be able to reach it. We set an address of 192.168.1.254/24, which Is still in our network. Now our computer still doesn't have an ip address but is assigned to use dhcp, so we needed to create a dhcp pool on the vlan to let the firewall know the usable IP addresses of the PC. We set the default gateway as 192.168.1.254 because that's the ip address of the vlan, a subnet of 255.255.255.0, and an inherited Primary/secondary DNS, primary secondary wins, primary nis, secondary nis, pop3 server, smtp server, and no dns suffix.

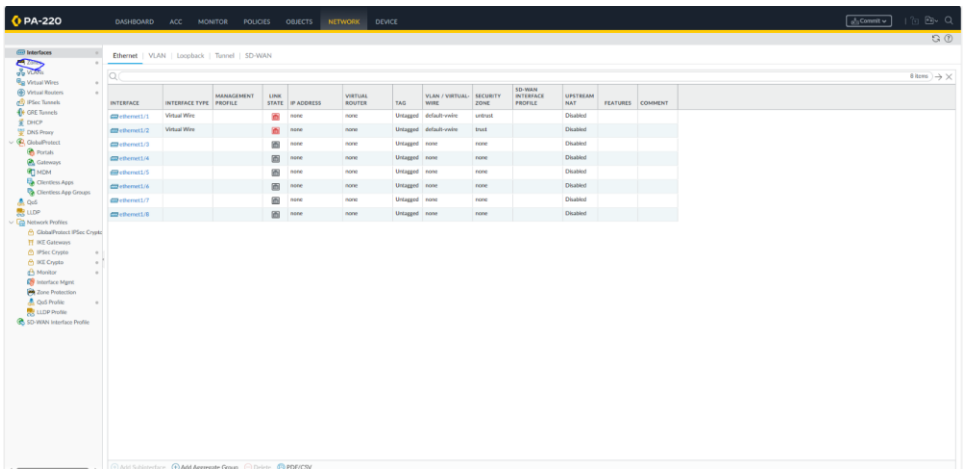
Lab Commands



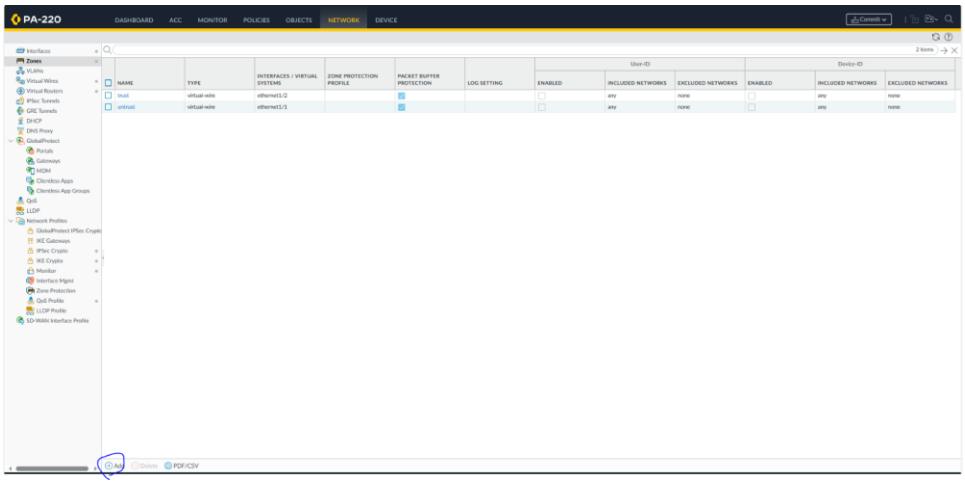
Enter the Web GUI



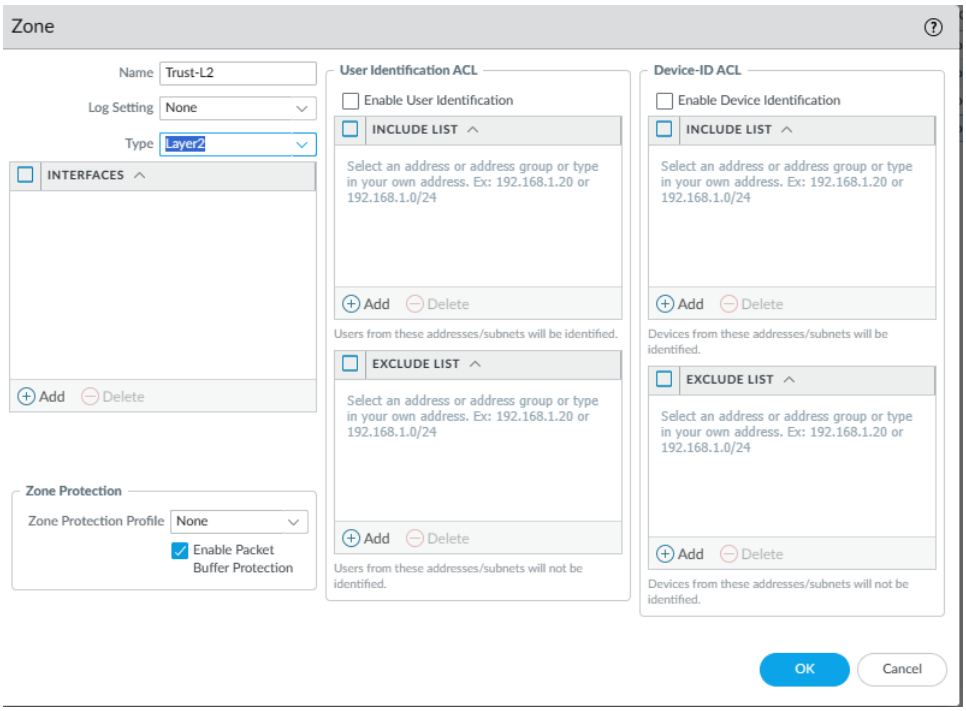
Go in the Network Tab on the top of the page



Click Zones in the left-hand toolbar



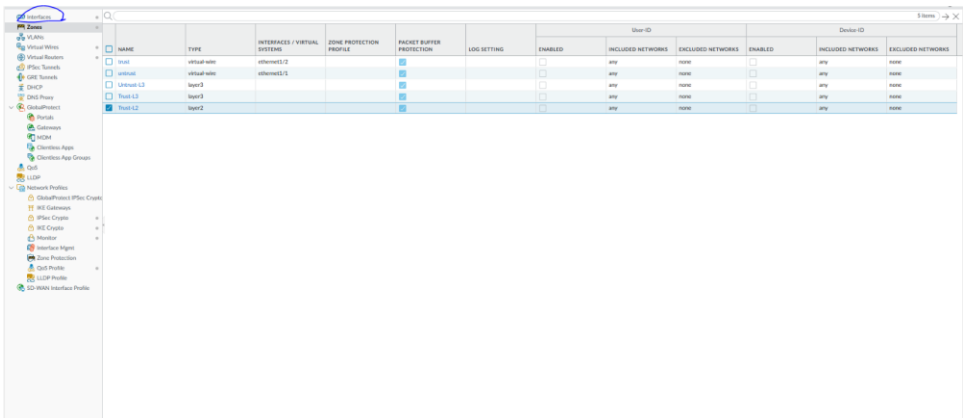
Add a zone by clicking add on the bottom left-hand corner



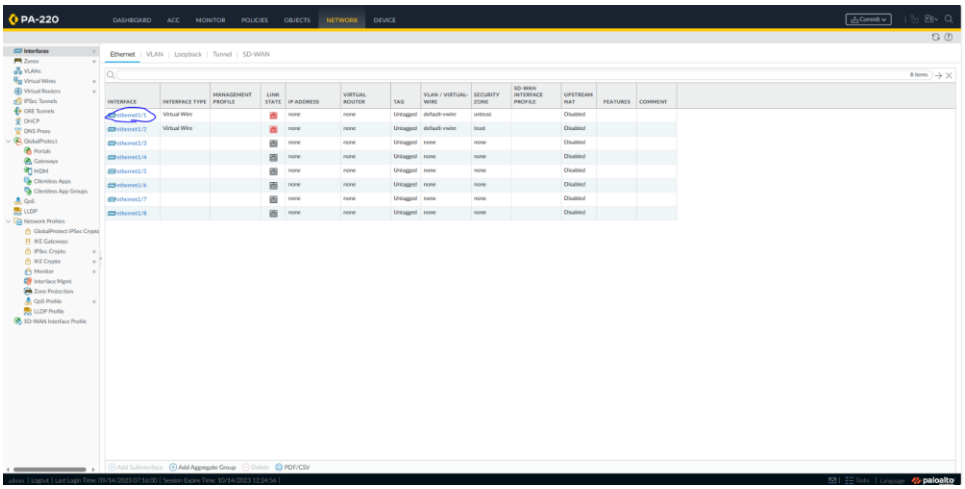
Create 3 zones, name the first zone Untrust-L3 in the name box. Click the drop-down type box and click layer3

Name the second zone Trust—L3 and click the drop-down type box and click layer 3

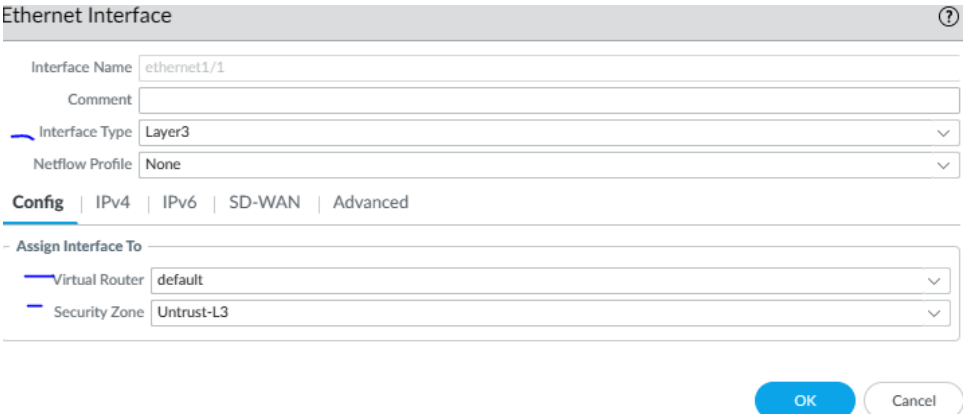
Name the third zone Trust-L2 and click the drop down type box and click layer 2



Enter Network > interfaces



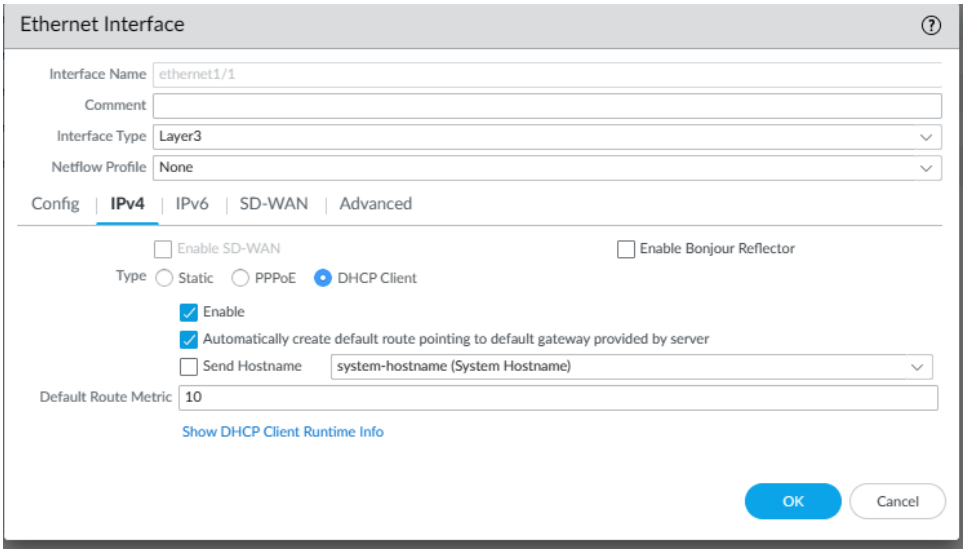
Click ethernet1/1 on the interface column



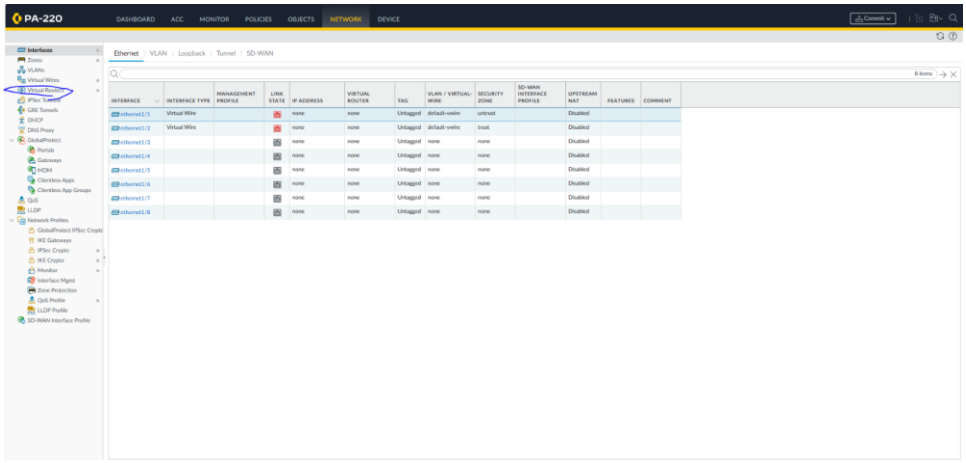
Change interface type to layer3

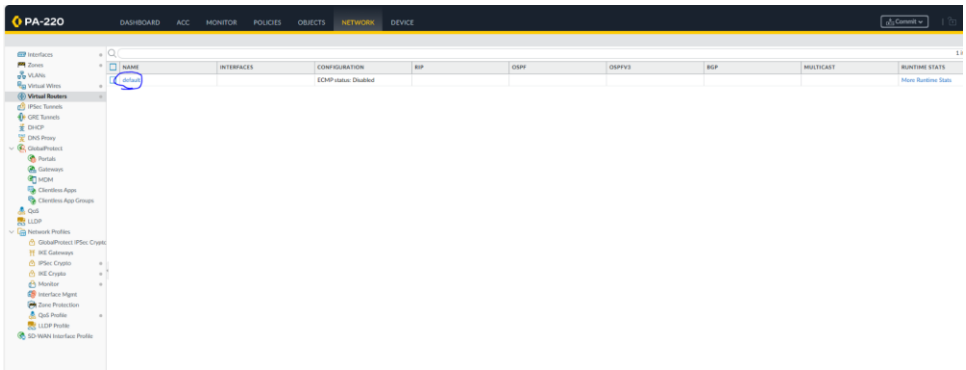
Change virtual router to default

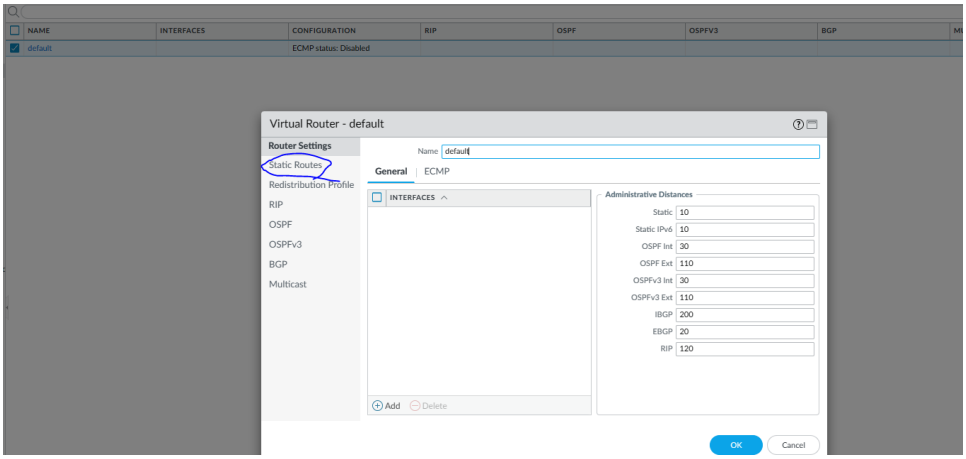
Change security zone to Untrust-L3

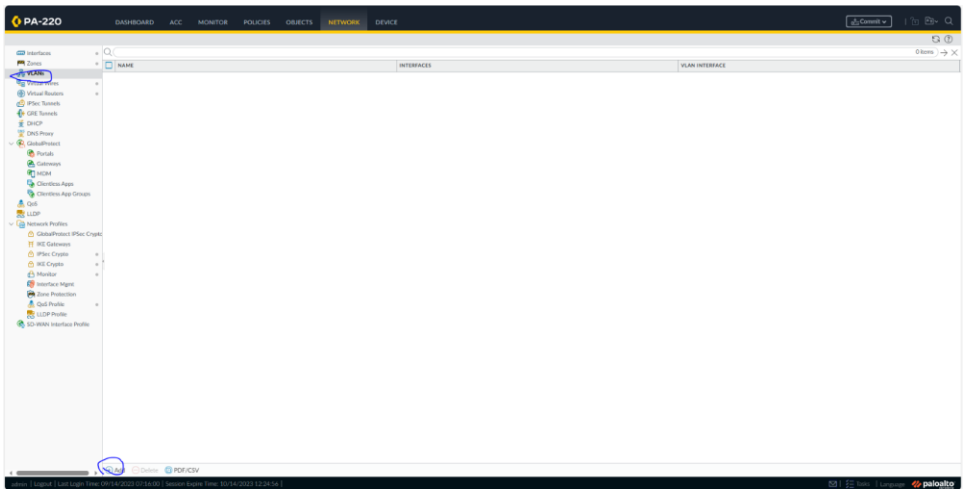


Click IPv4 tab in the ethernet interface. Then click on the DHCP client type.

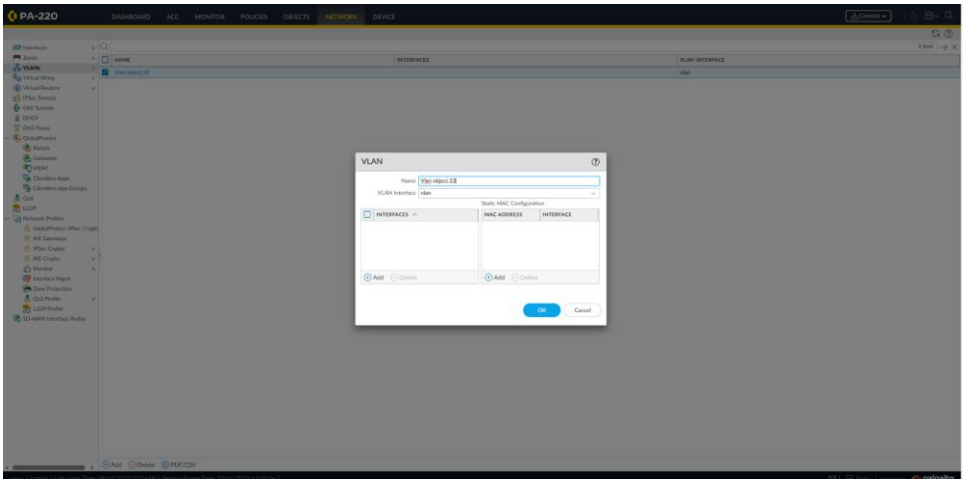


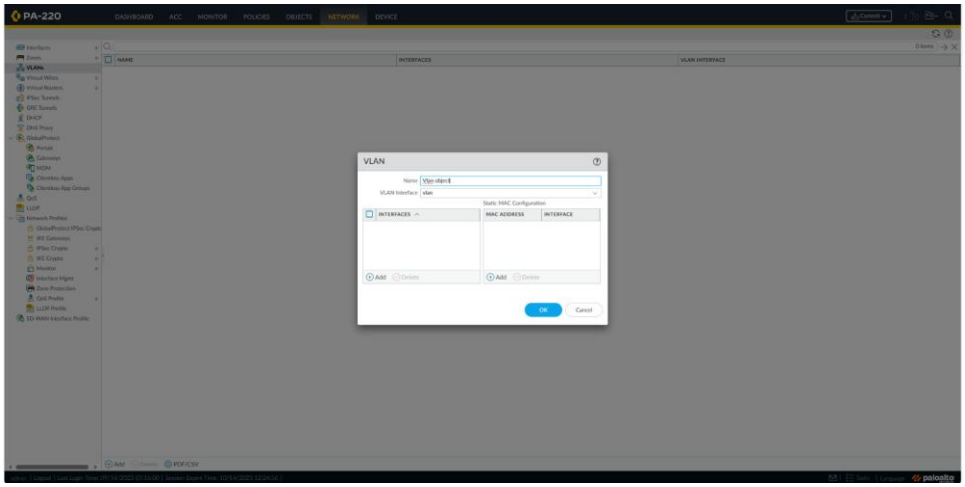




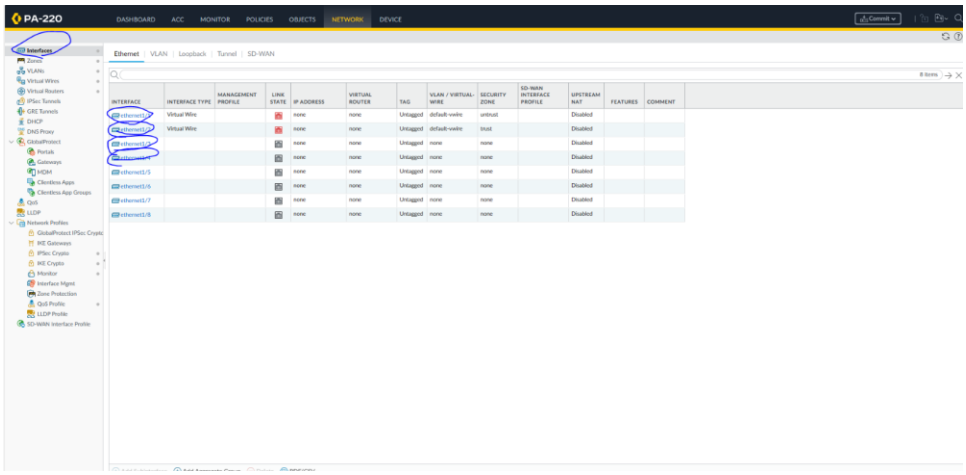


Join Network > Vlan > Add (in the bottom left corner)





Write “Vlan object.10” in the name box then press OK



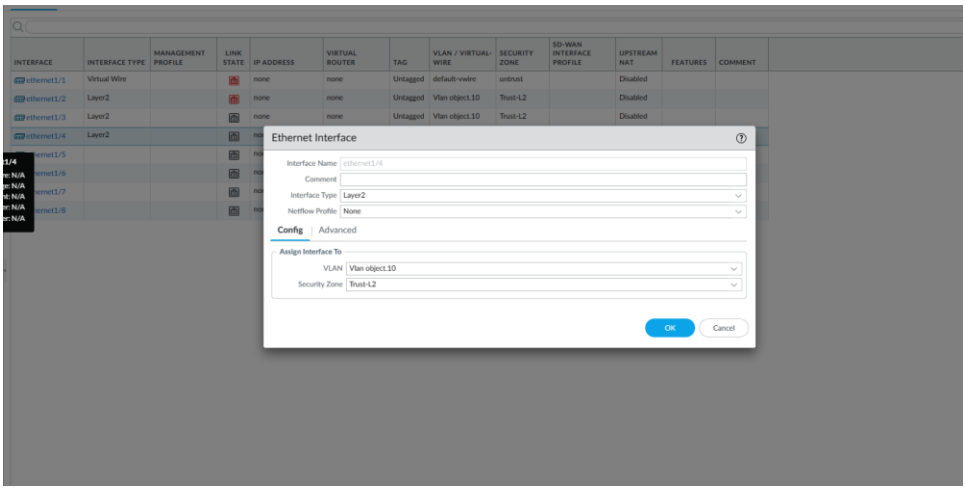
For this next step, ethernet 1/1- ethernet1/4 will all be configured with the same configuration

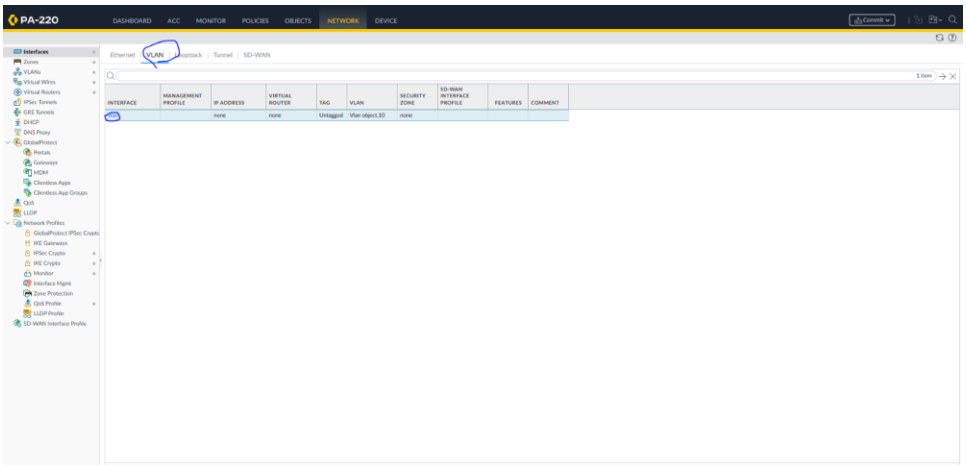
Set the interface type to layer 2

Set the netflow profile to none

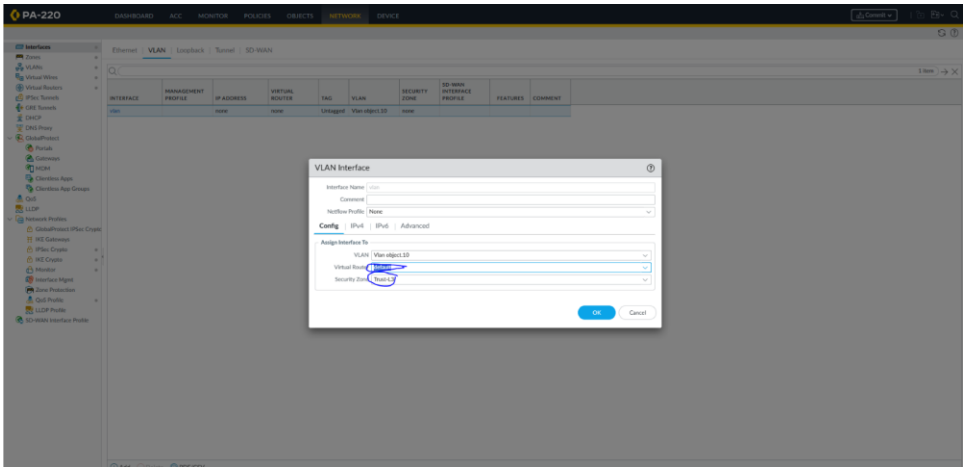
Set vlan to the Vlan object.10

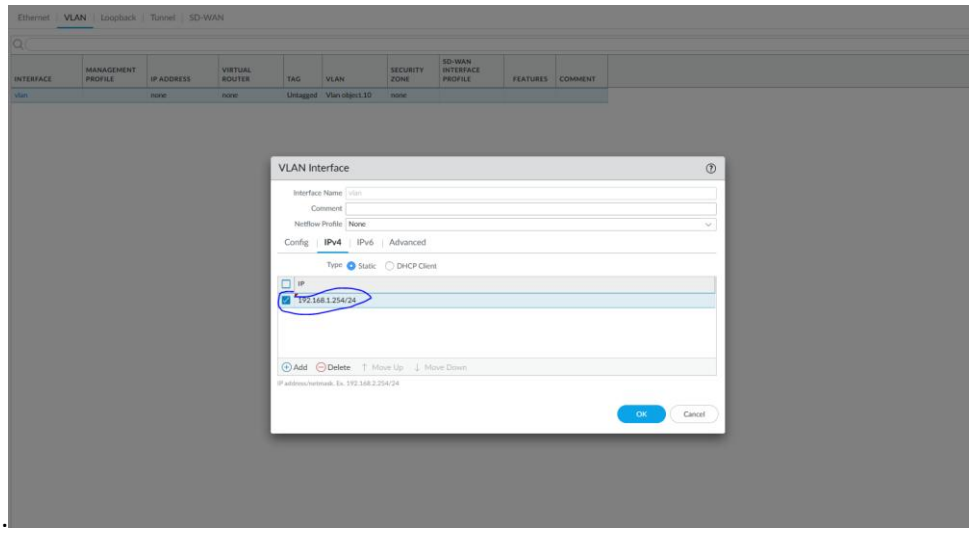
Set security zone to Trust-L2





Go back to interfaces > vlan and open “vlan”





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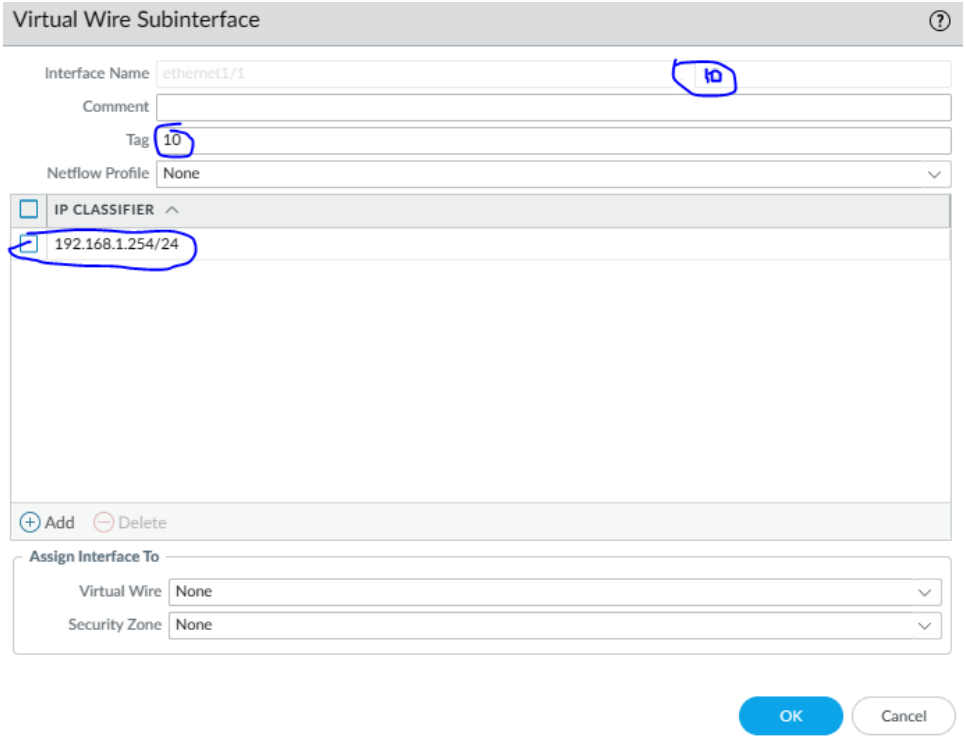
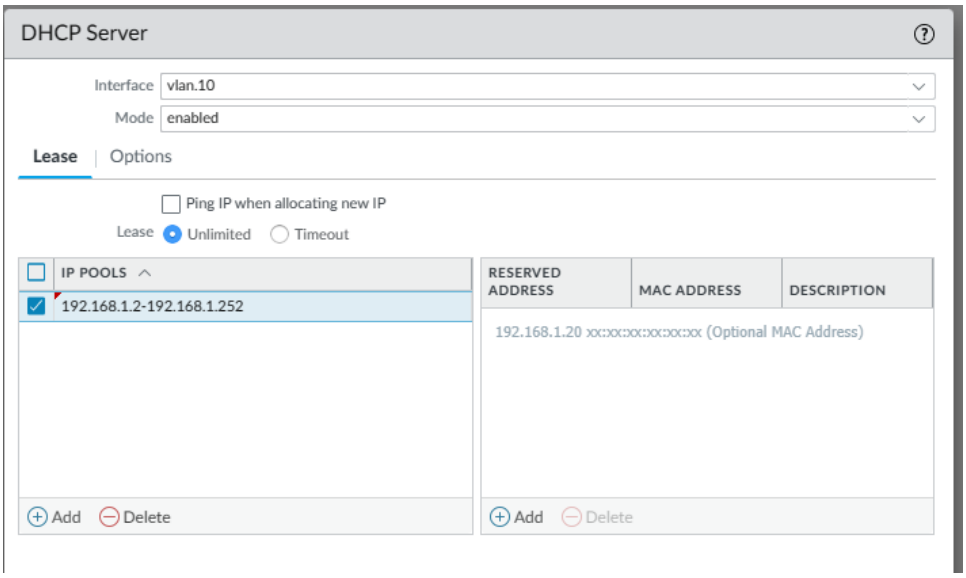
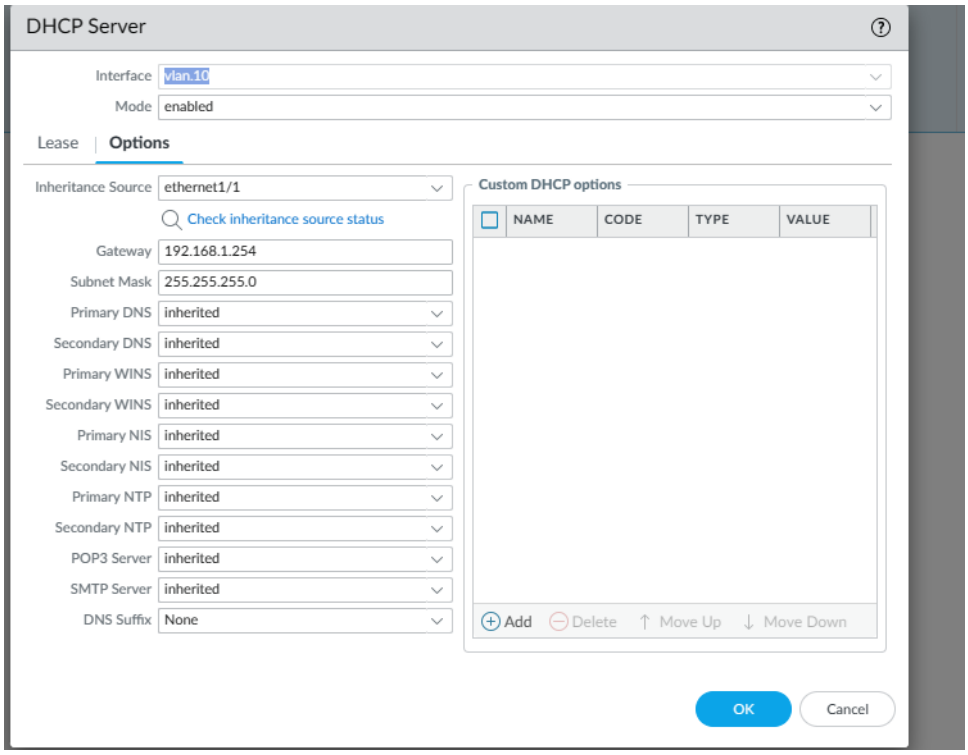
Select add and create an ip address of 192.168.1.254/24 set the vlan ID to 10



Enter Network > dhcp > and select add. Configure the interface with your selected vlan, select mode enabled.

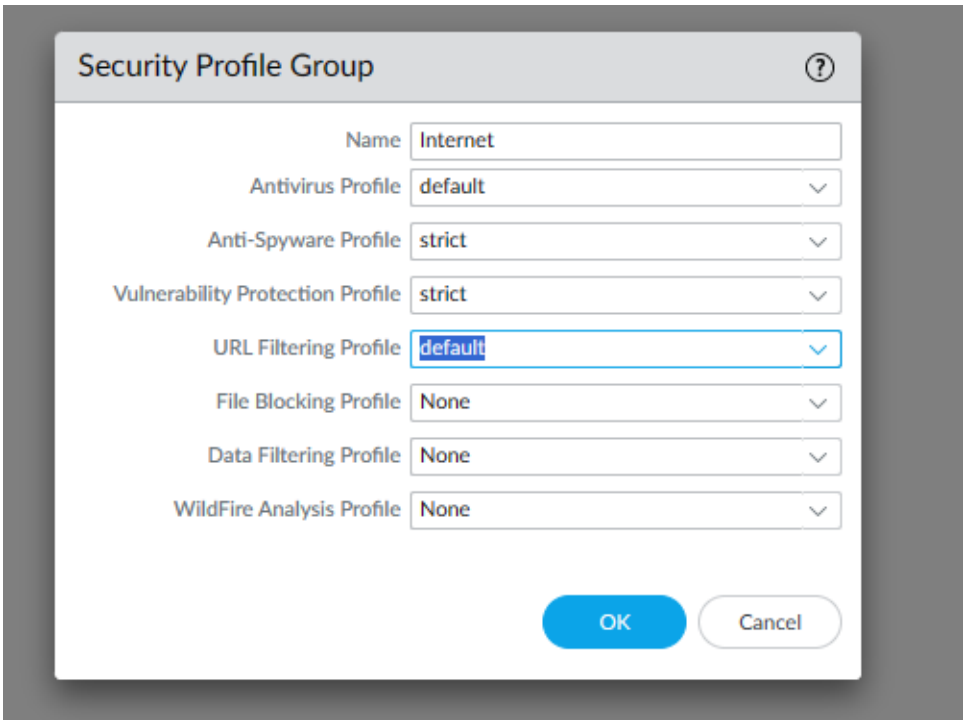


Select options and set the inheritance source as ethernet 1/1, the gateway to the ip you set two steps ago, a subnet mask of 255.255.255.0, and make everything else as inherited. The DNS suffix in this case is none as we are not configuring DNS.

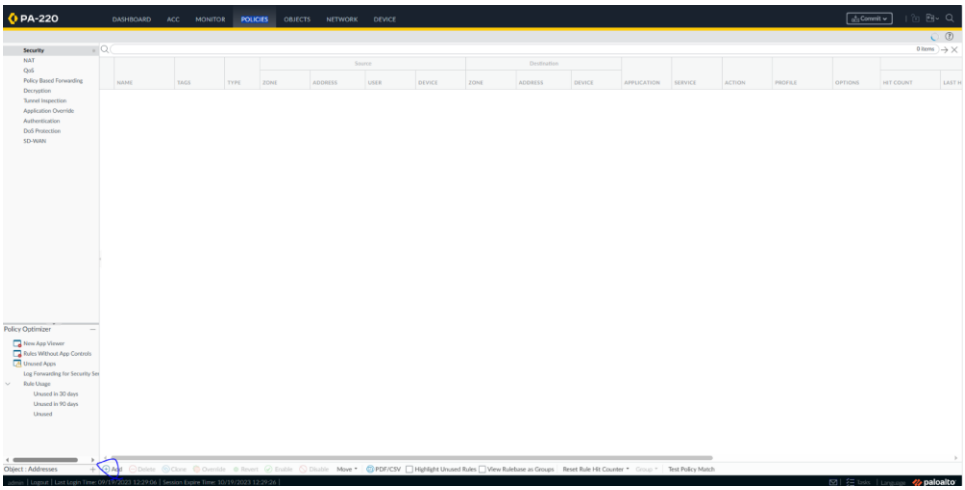


Enter >security profile groups

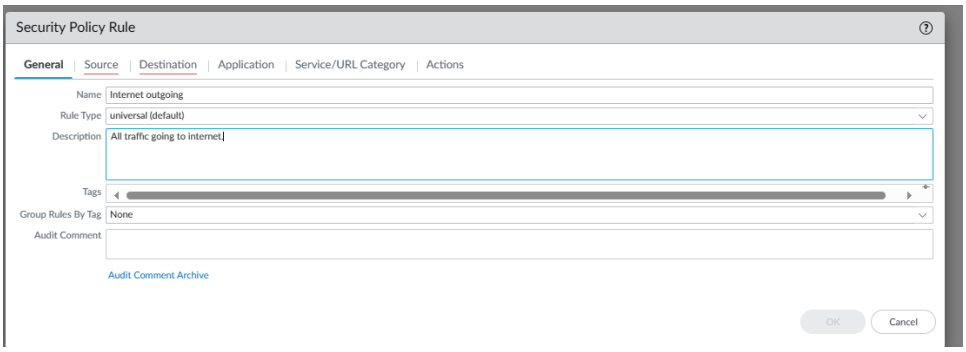
Select “add” and create a security profile group with the name “internet”, antivirus profile as default, anti spyware profile as strict, vulnerability protection profile as strict, and url filtering profile as default.



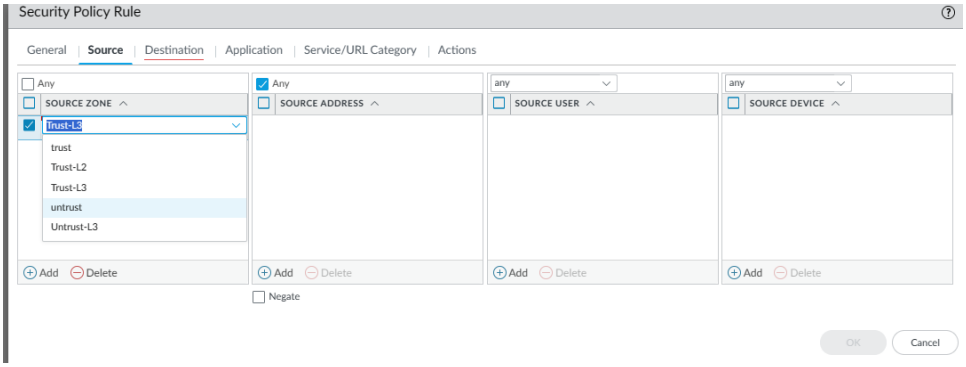
Enter Policies < security and select “add”.



Enter general and create a name of “internet outgoing” with the default rule type and a description .



Enter Source, add, and select trust-L3.



Enter destination and add Distrust-L3

Enter actions and change the profile type to “group”. After this, a new tab called group profile will open. Change the setting to Internet.

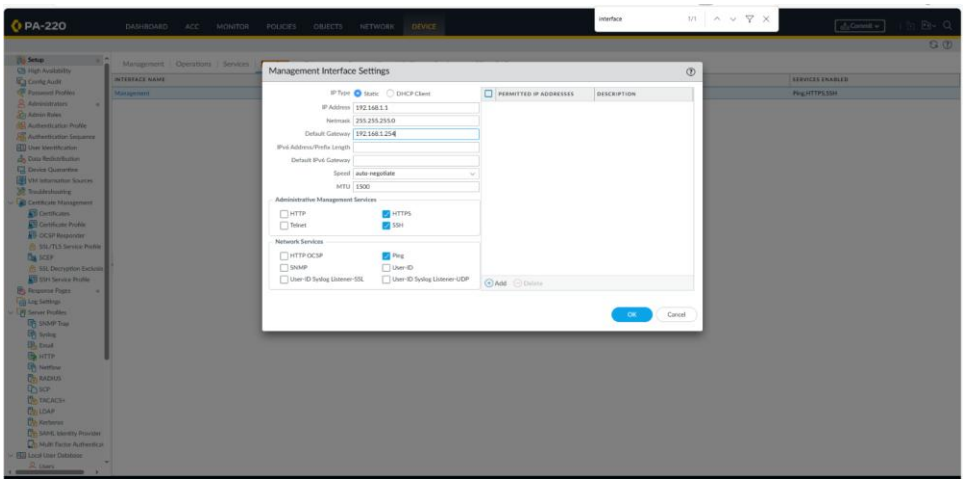
Select policies > Nat and select add.

Set the name to a name of your choosing and select ipv4 under NAT type.

Select the Original Packet tab and create a new source zone as Trust-L3, a destination zone of Untrust-L3, and a destination interface of eth1/1.

Enter the translated packet tab and change the translation type as dynamic ip and port, the address type as the interface address, and use the interface as eth1/1.

Enter Device > setup > management > management interface settings and change the default-gateway as your previously set default-gateway.



Network diagrams with IP's

The VLAN assigned to eth1-3 is configured with an IP of 192.168.1.254/24.

Configuration

Interfaces

Eth1/1

Interface Type: Layer 3

IP Address: Dynamic DHCP Client

Virtual Router: Default

Tag: Untagged

Vlan/Virtual Wire: None

Security zone: Untrusted

Eth1/2

Interface Type: Layer 2

IP Address: None

Virtual Router: Default

Tag: Untagged

Vlan/Virtual Wire: VLANMAN

Security zone: Trust-L2

Eth1/3

Interface Type: Layer 2

IP Address: None

Virtual Router: Default

Tag: Untagged

Vlan/Virtual Wire: VLANMAN

Security zone: Trust-L2

Eth1/4

Interface Type: Layer 2

IP Address: None

Virtual Router: Default

Tag: Untagged

Vlan/Virtual Wire: VLANMAN

Security zone: Trust-L2

VLANs

VLAN.10(vlan interface)

Netflow profile: None

VLAN: VLANMAN

Virtual router: Default

Security zone: Trust-L3

IPv4: 192.168.1.254/24

VLANMAN

Vlan interface: vlan.10

Interfaces: eth1/2, eth1/3, eth1/4

Problems

Vlan DNS issue

When the manually configured vlan on the firewall was not set to use a vlan ID other than 1, when we reached the step to configure a vlan, it told us that a DNS configuration was missing and caused us to restart.

Configuring random options after reaching the VLAN DNS issue

When we reached the vlan DNS issue, I started trying to fix it by setting random values for the DNS server/provider. After doing that, we couldn't really revert the changes so eventually decided to just factory reset the firewall again.

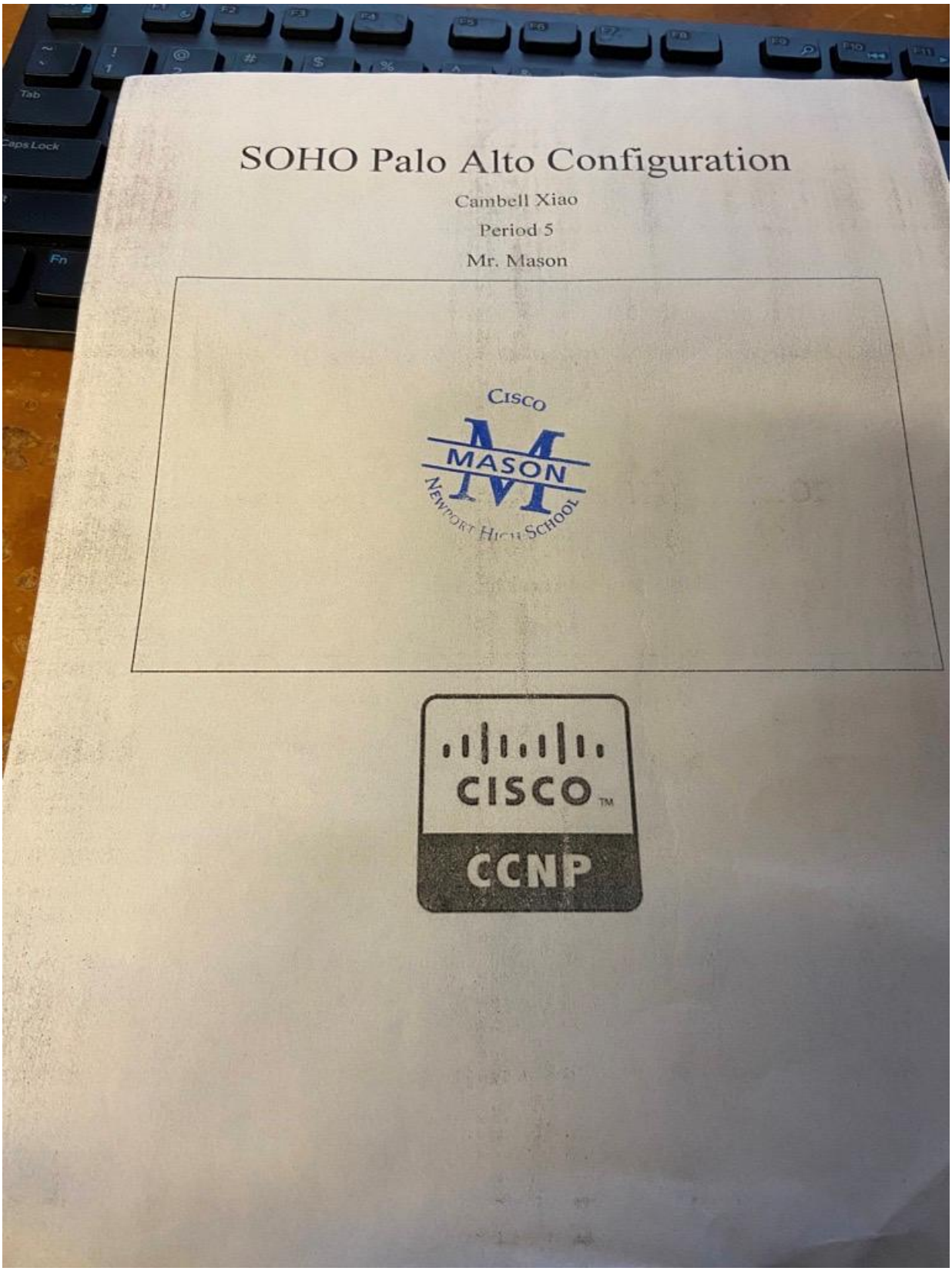
Misconfiguration

When we realized we were behind after factory resetting it, I started configuring everything quickly which led to one or two misconfigurations, most of which were in DHCP, which I had to go back and fix.

Conclusion

Configuring a palo alto firewall as the intermediary between your soho network and the internet can be very beneficial towards your network. While it may seem complicated and tedious to configure each firewall, it is very much worth it as it will protect not just your PC, but your company network from malicious code and filtering out unnecessary or malicious traffic.

Sign off


SOHO Palo Alto Configuration
Cambell Xiao
Period 5
Mr. Mason
CISCO
MASON
NEWPORT HIGH SCHOOL
CISCO
CCNP